

Compte Rendu LAB N°02 : Création et Gestion des conteneurs sous Docker

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Module : Cloud computing et Virtualisation

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Environnement : Docker ,NGINX



Objectifs du TP

- Maîtriser les commandes de base de Docker
- Créer et configurer des conteneurs avec limitation de ressources
- Gérer les réseaux Docker
- Sauvegarder, restaurer et cloner des conteneurs
- Mettre en place un serveur web simple (Nginx)

I. Création et configuration d'un conteneur Ubuntu

1. Téléchargement de l'image de référence

```
docker pull ubuntu:latest
```

Vérification :

```
docker images
```

```
PS C:\> docker pull ubuntu:latest
latest: Pulling from library/ubuntu
01d7766a2e4a: Pull complete
fd8cda969ed2: Download complete
Digest: sha256:d1e2e92c075e5ca139d51a140fff46f84315c0fdce203eab2807c7e495eff4f9
Status: Downloaded newer image for ubuntu:latest
docker.io/library/ubuntu:latest
PS C:\> docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	In Use
bde2020/hadoop-datanode:2.0.0-hadoop3.2.1-java8	ddf6e9ad55af	2.05GB	565MB	U
bde2020/hadoop-historyserver:2.0.0-hadoop3.2.1-java8	ab2971d5f8c3	2.05GB	565MB	U
bde2020/hadoop-namenode:2.0.0-hadoop3.2.1-java8	51ad9293ec52	2.05GB	565MB	U
bde2020/hadoop-nodemanager:2.0.0-hadoop3.2.1-java8	b5515b537a30	2.05GB	565MB	U
bde2020/hadoop-resourcemanager:2.0.0-hadoop3.2.1-java8	91ca2afe8616	2.05GB	565MB	U
ubuntu:latest	d1e2e92c075e	119MB	31.7MB	

```
PS C:\> |
```

2. Création du conteneur principal avec contraintes

```
docker run -it `
  --name ubuntu_youssef `
  --cpus="1.0" `
  --memory="512m" `
  --hostname ubuntu-youssef `
  ubuntu:latest bash
```

```
PS C:\> docker run -it --name ubuntu_youssef --cpus="1.0" --memory="512m"
--hostname ubuntu-youssef ubuntu:latest /bin/bash
root@ubuntu-youssef:/# |
```

3. Vérification de l'état des conteneurs

```
docker ps -a
```

```
PS C:\> docker ps -a
CONTAINER ID   IMAGE                                COMMAND                  CREATED          STATUS          PORTS
78a78c219004   ubuntu:latest                       "/bin/bash"             56 seconds ago  Exited (127)    8 seconds ago
1be44c8f8adb   bde2020/hadoop-nodemanager:2.0.0-hadoop3.2.1-java8 /entrypoint.sh /run...  8 days ago      Exited (255)    24 hours ago
```

```
docker inspect ubuntu_youssef
```

```
PS C:\> docker inspect ubuntu_youssef
[
  {
    "Id": "78a78c219004431bb2d3b5f3a06b4154eac9d6320a4ef211a1da6db96b83983f",
    "Created": "2026-02-26T09:31:15.821476073Z",
    "Path": "/bin/bash",
    "Args": [],
    "State": {
      "Status": "exited",
      "Running": false,
      "Paused": false,
      "Restarting": false,
      "OOMKilled": false,
      "Dead": false,
      "Pid": 0,
      "ExitCode": 127,
      "Error": "",
      "StartedAt": "2026-02-26T09:31:16.286901404Z",
      "FinishedAt": "2026-02-26T09:32:02.92551762Z"
    },
    "Image": "sha256:d1e2e92c075e5ca139d51a140fff46f84315c0fdce203eab"
```

4. Création et connexion à un réseau personnalisé

```
docker network create ubuntu_network
```

```
docker network connect ubuntu_network ubuntu_youssef
```

```
docker network inspect ubuntu_network
```

```

PS C:\> docker network create ubuntu_network
1ccf5351c368022a2e07b3e9dad8c563dd3ba3bccbce0d788771a6331bc9ce7d
PS C:\> docker network connect ubuntu_network ubuntu_youssef
PS C:\> docker network inspect ubuntu_network
[
  {
    "Name": "ubuntu_network",
    "Id": "1ccf5351c368022a2e07b3e9dad8c563dd3ba3bccbce0d788771a6331bc9ce7d",
    "Created": "2026-02-26T09:35:02.970348758Z",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": {},
      "Config": [
        {
          "Subnet": "172.19.0.0/16",
          "Gateway": "172.19.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Options": {
      "com.docker.network.enable_ipv4": "true",
      "com.docker.network.enable_ipv6": "false"
    }
  }
]

```

6. Accès au conteneur

`docker exec -it ubuntu_lab bash`

```

]
PS C:\> docker start ubuntu_youssef
ubuntu_youssef
PS C:\> docker exec -it ubuntu_youssef /bin/bash
root@ubuntu-youssef:/# |

```

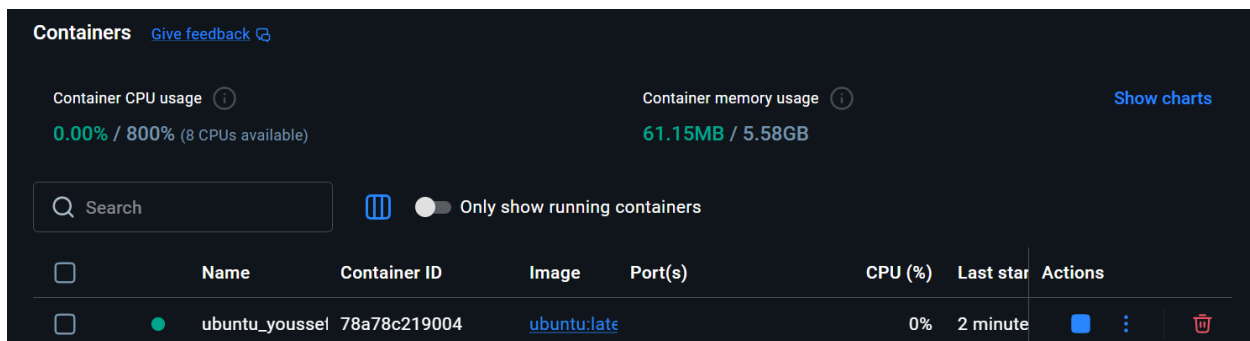
7. Mise à jour du système (dans le conteneur)

`apt update && apt upgrade -y`

```

root@ubuntu-youssef:/# apt update
Get:1 http://archive.ubuntu.com/ubuntu noble InRelease [256 kB]
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:3 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1210 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble/main amd64 Packages [1808 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1879 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble/restricted amd64 Packages [117 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [331 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble/universe amd64 Packages [19.3 MB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [3246 kB]
Get:12 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [34.8 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [2276 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [

```



8. Installation d'outils de base

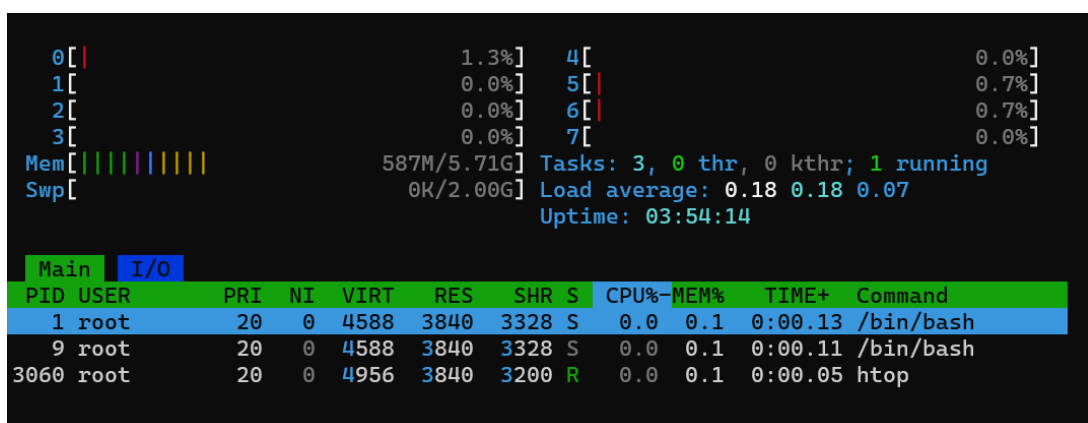
```
apt install nano net-tools curl htop -y
```

```

root@ubuntu-youssef:/# apt install nano net-tools curl htop -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
nano is already the newest version (7.2-2ubuntu0.1).
net-tools is already the newest version (2.10-0.1ubuntu4.4).
curl is already the newest version (8.5.0-2ubuntu10.7).
htop is already the newest version (3.3.0-4build1).

```

Tests :



9. Vérification de l'isolation

Création d'un second conteneur :

```
docker run -it --name ubuntu_test ubuntu:latest bash
```

Dans ubuntu_youssef :

```
PS C:\> docker exec -it ubuntu_youssef bash
root@ubuntu-youssef:/# ls
bin          etc          media        root         srv          var
bin.usr-is-merged  home        mnt          run          sys
boot         lib          opt          sbin         tmp
dev          lib64        proc         sbin.usr-is-merged  usr
root@ubuntu-youssef:/#
```

```
touch /root/fichier_lab.txt
```

Dans ubuntu_test :

```
ls
```

```
PS C:\Users\youssef hachimi> docker run -it --name ubuntu2 ubuntu:latest /bin/bash
root@00bf4714fa6a:/# touch fichier_lab.txt
root@00bf4714fa6a:/# ls
bin      etc      lib      mnt      root      srv      usr
boot     fichier_lab.txt  lib64    opt      run      sys      var
dev      home     media    proc     sbin     tmp
root@00bf4714fa6a:/#
```

→ fichier_lab.txt n'apparaît PAS → isolation confirmée

II. Sauvegarde, restauration et clonage

10. Création d'une image de sauvegarde (commit)

```
docker commit ubuntu_lab ubuntu_backup_image
```

```
PS C:\> docker commit ubuntu_youssef ubuntu_backup_image
sha256:a5021a6fd1ac466fbe3cacd19b31a1ce946c7efabb76dba401b17c4ca3732f7c
```

Export au format tar :

```
docker save -o .\ubuntu_backup.tar ubuntu_backup_image
```

```
PS C:\> docker save -o .\ubuntu_backup.tar ubuntu_backup_image
PS C:\> ls
```

Directory: C:\

Mode	LastWriteTime		Length	Name
----	-----	-----	-----	----
d-----	2/18/2026	9:24 AM		hadoop_docker
d-----	11/26/2025	1:16 AM		inetpub
d-----	2/26/2026	9:45 AM		Intel
d-----	11/26/2025	1:16 AM		PerfLogs
d-----	2/25/2026	11:00 AM		Program Files
d-r---	2/9/2026	2:41 PM		Program Files (x86)
d-----	12/28/2025	5:11 PM		Python314
d-----	11/29/2025	3:28 PM		SWSetup
d-----	2/21/2026	11:10 PM		test
d-r---	2/21/2026	9:15 AM		Users
d-----	2/21/2026	12:59 AM		Windows
-a----	2/26/2026	10:02 AM	77361664	ubuntu_backup.tar

```
PS C:\>
```

11. Restauration depuis la sauvegarde

```
docker load -i .\ubuntu_backup.tar
```

```
docker run -it `
  --name ubuntu_restored `
  ubuntu_backup_image `
  bash
```

```
PS C:\> docker load -i ubuntu_backup.tar
Loaded image: ubuntu_backup_image:latest
PS C:\> docker run -it --name ubuntu_restored ubuntu_backup_image /bin/bash
root@f7da577c7d88:/#
```

12. Clonage simple

Docker start ubuntu_restored

```
PS C:\> docker start ubuntu_restored
ubuntu_restored
PS C:\> docker exec -it ubuntu_restored /bin/bash
root@f7da577c7d88:/# ls
bin          etc          media        root          srv          var
bin.usr-is-merged  home        mnt          run           sys
boot         lib          opt          sbin          tmp
dev          lib64        proc         sbin.usr-is-merged  usr
root@f7da577c7d88:/#
```

```
docker commit ubuntu_lab ubuntu_clone_image
```

```
PS C:\> docker commit ubuntu_youssef ubuntu_clone
sha256:34e0aa3a8b7e89558a4c0c93c817dbadef14f956d65000b6610f67ac1a8c0705
PS C:\> docker run -it --name ubuntu_clonee ubuntu_clone /bin/bash
root@16936e001c28:/#
```



```
docker run -it `
  --name ubuntu_clone `
  ubuntu_clone_image `
  bash
```

```
PS C:\> docker start ubuntu_clonee
ubuntu_clonee
PS C:\> docker exec -it ubuntu_clonee bash
root@16936e001c28:/# uname -a
Linux 16936e001c28 6.6.87.2-microsoft-standard-WSL2 #1 SMP PREEM
PT_DYNAMIC Thu Jun  5 18:30:46 UTC 2025 x86_64 x86_64 x86_64 GNU
/Linux
root@16936e001c28:/#
```

III. Surveillance des ressources

```
docker stats
```

```
PS C:\> docker stats
```

```
docker top ubuntu_youssef
```

CONTAINER ID	NAME	CPU %	MEM USAGE / LIMIT
16936e001c28	ubuntu_clonee	0.00%	928KiB / 5.709GiB
0.02%	872B / 126B	0B / 0B	1
f7da577c7d88	ubuntu_restored	0.00%	952KiB / 5.709GiB
0.02%	1.12kB / 126B	0B / 4.1kB	1
00bf4714fa6a	ubuntu2	0.00%	964KiB / 5.709GiB
0.02%	1.5kB / 126B	766kB / 0B	1
b053a37c3c56	ubuntu_test	0.00%	4.219MiB / 5.709GiB
0.07%	2.45kB / 126B	3.96MB / 0B	1
78a78c219004	ubuntu_youssef	0.00%	956KiB / 512MiB
0.18%	3.07kB / 252B	1.44MB / 4.1kB	1

```
PS C:\> docker top ubuntu_youssef
UID                PID                PPID                C                CMD
-----
root               527               505                 0                pts/0
09:58              /bin/bash
PS C:\> |
```

Sur Docker :

☐	☐	ubuntu_youssef	78a78c219004	ubuntu:late	0%	0B / 0B			
☐	☒	ubuntu_test	b053a37c3c56	ubuntu:late	0%	4.22MB /			
☐	☒	ubuntu2	00bf4714fa6a	ubuntu:late	0%	964KB /			
☐	☒	ubuntu_restore	f7da577c7d88	ubuntu:ba	0%	952KB /			
☐	☒	ubuntu_clonee	16936e001c28	ubuntu:clo	0%	928KB /			

IV. Déploiement d'un serveur web Nginx

1. Création du conteneur web

```
PS C:\> docker start ubuntu_youssef
ubuntu_youssef
PS C:\> docker exec -it ubuntu_youssef bash
root@ubuntu-youssef:/# apt update
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages
[2017 kB]
88% [5 Packages 1686 kB/2017 kB 84%]
```

```
docker run -it `
--name ubuntu_web `
-p 8080:80 `
ubuntu:latest bash
```

2. Mise à jour et installation de Nginx

(Dans le conteneur)

```
apt update && apt upgrade -y
```

```
apt install -y nginx
```

```
root@ubuntu-youssef:/# apt install nginx -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  iproute2 libatm1t64 libbbpf1 libcap2-bin libelf1t64 libmnl0 libpam-cap
  libxtables12 nginx-common
Suggested packages:
  iproute2-doc python3:any fcgiwrap nginx-doc ssl-cert
The following NEW packages will be installed:
  iproute2 libatm1t64 libbbpf1 libcap2-bin libelf1t64 libmnl0 libpam-cap
  libxtables12 nginx nginx-common
0 upgraded, 10 newly installed, 0 to remove and 0 not upgraded.
Need to get 2026 kB of archives.
After this operation, 5799 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libelf1t64 am
d64 0.190-1.1ubuntu0.1 [57.8 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/main amd64 libbbpf1 amd64 1:1.3.0
-2build2 [166 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble/main amd64 libmnl0 amd64 1.0.5-2
build1 [12.3 kB]
```

Vérification de la version de Nginx dans le conteneur de base
Vérification de la version installée de Nginx (1.24.0 – build
Ubuntu) directement dans le conteneur Ubuntu de base avant
création de l'image personnalisée.

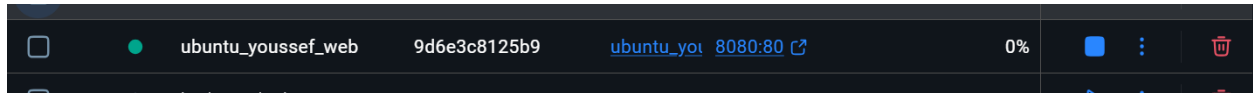
```
root@ubuntu-youssef:/# nginx -v
nginx version: nginx/1.24.0 (Ubuntu)
root@ubuntu-youssef:/#
```

□ **Création de l'image Docker à partir du conteneur modifié** Commit du conteneur actif ubuntu_youssef vers une nouvelle image nommée ubuntu_youssef_nginx (SHA256: 0992c144...505b7a). Cette étape fige l'état du conteneur après installation et configuration initiale.

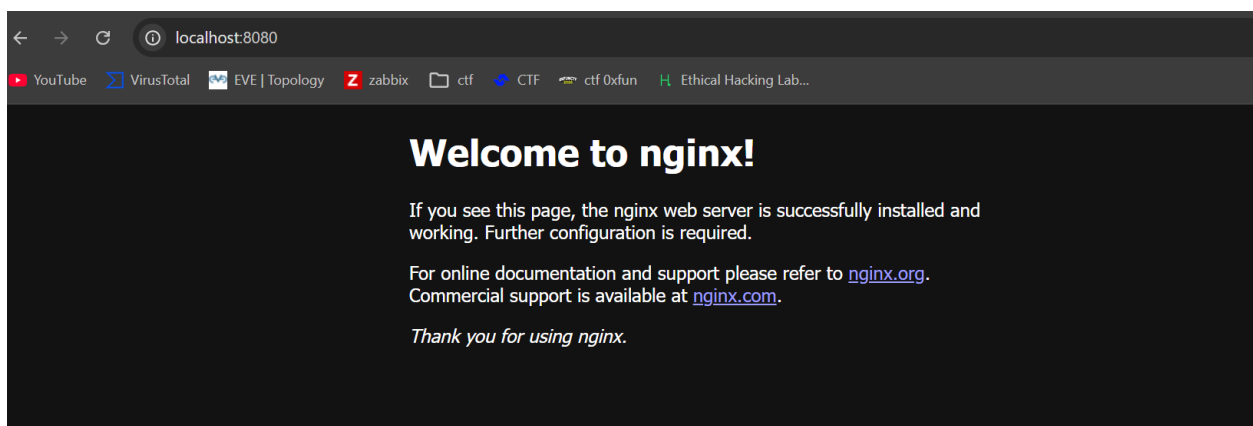
```
PS C:\Users\youssef hachimi> docker commit ubuntu_youssef ubuntu_youssef_nginx
sha256:0992c14429cb74a6f33f684701adaa43c995c2d6ad557dd42925c4a19f505b7a
PS C:\Users\youssef hachimi> docker run -d --name ubuntu_youssef_web -p 8080:80 ubuntu_youssef_nginx nginx -g "daemon off;"
9d6e3c8125b98f032c25cca08d182b938f216a02e654205832f77c07230fad3c
PS C:\Users\youssef hachimi>
```

▣ Vue Docker Desktop – conteneur en cours d'exécution

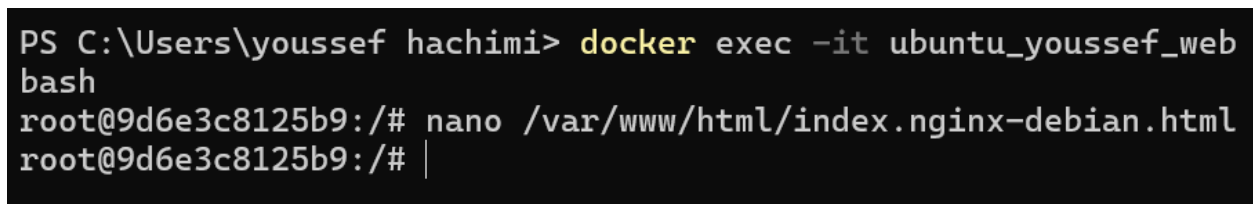
l'interface Docker Desktop montrant le conteneur ubuntu_youssef_web actif, avec le port 8080:80 mappé



– page Nginx par défaut visible dans le navigateur Accès réussi à `http://localhost:8080` affichant la page d'accueil par défaut de Nginx (« Welcome to nginx! »), confirmant que le serveur web est opérationnel et accessible depuis l'hôte.



▣ Modification du fichier `index.html` par défaut via nano Accès interactif au conteneur en cours via `docker exec -it` et ouverture du fichier `/var/www/html/index.nginx-debian.html` avec nano pour personnalisation de la page d'accueil.



Après Modification sur le code source de page web :

